

ENTERPRISE CRM TARGET MODEL

Binding Implementation Specification

CRM + Customer Need Intelligence + Configurable Pipeline + Sales Quote Portal

Status: BINDING ARCHITECTURE DECISIONS

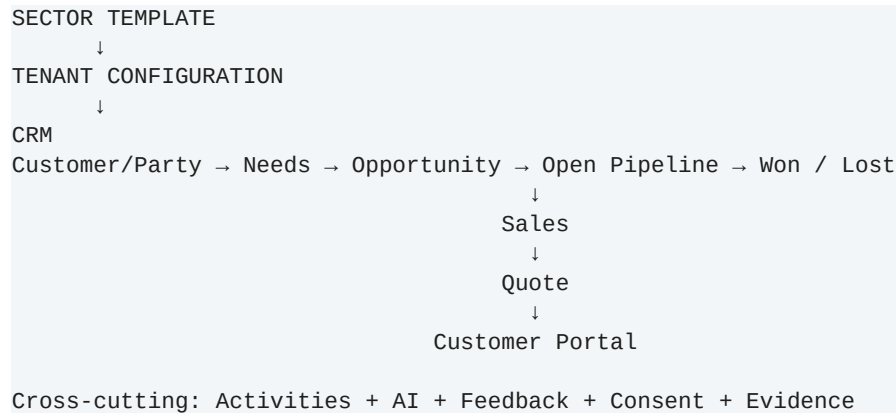
Date: 2026-09-16

Target implementation: .NET 10 backend, React frontend

Purpose: This document is intended to be given directly to Claude/Codex as the authoritative target model. The implementation must follow the frozen decisions and guardrails in this document. Where the current code differs, the agent must propose a migration/delta plan before changing code.

1. Executive Summary

The target CRM is not a fixed sales funnel. It is a platform with a stable lifecycle, a tenant-configurable pipeline, a first-class customer-need model, and a separate Sales quote/document flow. AI assists the process by inferring needs and context, but AI output is not automatically business truth.



Core rule: CRM owns the relationship, need, opportunity, pipeline and activity context. Sales owns quotes and formal commercial documents. Privacy/Consent owns permissions. Feedback owns evaluations. AI can suggest and automate within policy, but it does not redefine domain truth.

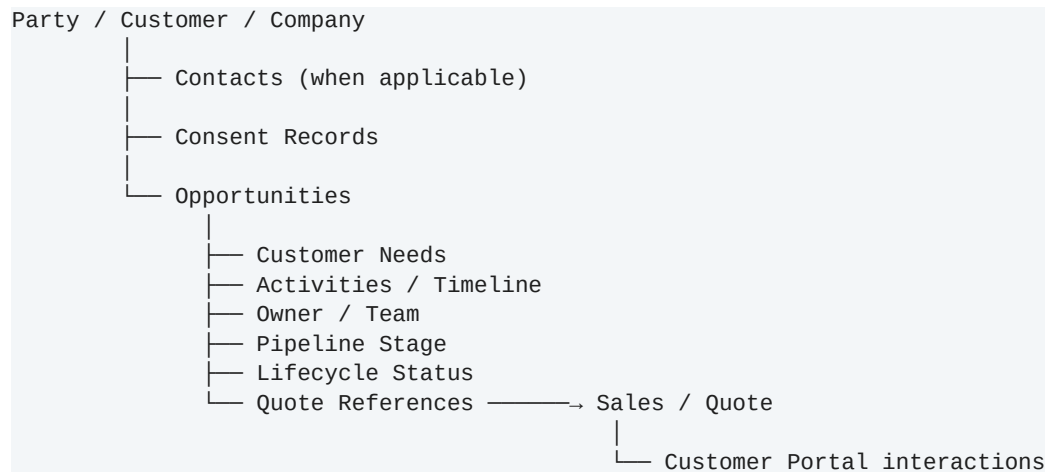
2. Frozen Decisions - Do Not Reinterpret

Decision	Binding rule
Lifecycle	Exactly: Draft, Open, Won, Lost. These are platform-level canonical states and are not tenant-editable.
Pipeline	Tenant/sector configurable. Pipeline stages describe the Open phase. Names, number, order and allowed actions may vary.
Quote timing	A quote may be created/sent during any Open stage where tenant policy allows it. Quote is not tied to a hardcoded stage name.
Quote ownership	Quote belongs to Sales, not CRM. Opportunity may reference one or more Quotes.
Quote status	Draft/Sent/Viewed/Accepted/Rejected/Expired are quote states, not Opportunity lifecycle states.
Won	Commercial opportunity is closed positively. Formal downstream documents may be produced by Sales/Document/Finance flows.
Lost	Commercial opportunity is closed negatively. No normal forward-selling action occurs on that closed opportunity.
Customer Need	First-class CRM concept. It represents the customer's detected/declared needs and an approximate economic potential, not guaranteed revenue.
AI	AI may infer needs from voice, notes and product interaction. AI

	inference must preserve source and confidence. It must not silently become confirmed business truth.
Sector + tenant	Sector templates provide defaults; tenant configuration may override them without changing the canonical lifecycle.

3. Target Conceptual Model

The model supports both simple retail and complex B2B processes without forcing every tenant to use every capability.



Concept	Meaning	Required?
Party / Customer	Person or organization with whom the tenant has a commercial relationship.	Yes
Contact	Human contact related to an organization/account.	Conditional
Lead	Pre-opportunity prospect/pursuit. Useful in some sectors, unnecessary in others.	Optional capability
Opportunity	Commercial pursuit that may create revenue.	Yes
Customer Need	Need detected or declared for this party/opportunity.	Yes when tenant uses need intelligence
Pipeline Stage	Tenant-specific position inside Open.	Yes for active opportunities; at least one Open stage
Activity	Call, meeting, note, task, email, quote interaction, etc.	Yes for CRM usefulness
Quote	Commercial offer document and its revisions.	Sales domain
Feedback	Employee/branch/tenant/service evaluations.	Separate capability

Consent	Purpose-specific permission/evidence record.	Separate privacy capability
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4. Canonical Opportunity Lifecycle



Default: Won and Lost are terminal.

Reopen requires an explicit command + reason + policy + evidence.

Lifecycle	Meaning	Typical allowed actions
Draft	Record is being prepared; not yet an active sales pursuit.	Edit basic data, add needs, assign owner, discard, open.
Open	Active revenue pursuit.	Change pipeline stage, add activities/needs, create/send quote if policy allows, negotiate, win/loss close.
Won	Positive commercial close.	Read/history, formal downstream documents, feedback, after-sales handoff. No new quote by default.
Lost	Negative commercial close.	Loss reason, notes, feedback, follow-up/new opportunity. No normal new quote.

Analytics rule: Draft records that never entered Open must not distort win/loss conversion metrics. OpenedAt/EnteredOpenAt must be available so pipeline conversion can use only opportunities that actually entered the active funnel.

5. Configurable Pipeline Model

Pipeline stages are tenant-facing business steps inside the Open lifecycle. They are not canonical lifecycle states.

Platform lifecycle: Open

Tenant A - Furniture

Need Analysis → Quote Given → Negotiation

Tenant B - Technical Service

Request Received → Site Visit → Proposal → Approval Waiting

Tenant C - Simple Retail

Waiting

All three still use the same canonical lifecycle: Draft / Open / Won / Lost.

Pipeline field	Rule
PipelineDefinition	Tenant-scoped definition; may originate from a sector template.
Version	Pipeline changes create a new version. Existing open opportunities are not silently remapped.
Stage	Tenant-defined name, order, active flag and allowed capabilities.
Allowed actions	Stage may allow/restrict capabilities such as quote.create, quote.send, discount.request, approval.request.
Terminal outcome	Won/Lost are lifecycle transitions, not ordinary configurable Open stages.
Minimum complexity	A tenant may use only one Open stage.
Renaming	Tenant may rename any Open stage without changing lifecycle meaning.

Do not hardcode: Never write code that checks stage name strings such as "Teklif Verildi" or "Negotiation". Business behavior must use stable stage IDs and/or capability/policy keys.

6. Customer Need Intelligence

Customer Need is a first-class CRM capability. It estimates the economic potential of the customer's needs even when exact purchase intent is not yet known.

Furniture example

Need catalog reference values:

Bedroom Set 120,000 TL

Dining Room 90,000 TL

Mattress 30,000 TL

Detected customer needs:

Bedroom Set + Dining Room + Mattress

Estimated Need Potential ≈ 240,000 TL

This is NOT a quote total and NOT guaranteed revenue.

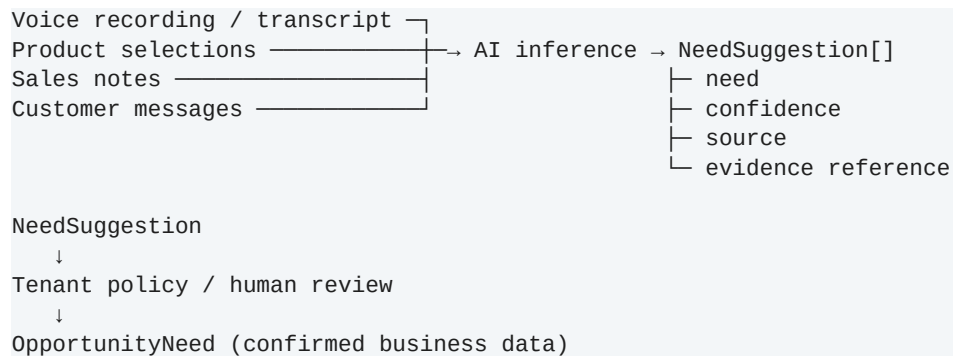
Object / field	Meaning
CustomerNeedDefinition	Tenant/sector need catalog item, e.g. Bedroom Set, Annual Maintenance, Legal Consultancy.
ReferenceValue	Approximate average economic value used for potential estimation.
OpportunityNeed	Need associated with a specific opportunity.
EstimatedValueSnapshot	Value copied/snapshotted when the need is attached so historical reporting does not change when the catalog reference

	changes.
Source	manual, ai_voice, ai_product_inference, ai_note, integration, etc.
Confidence	AI confidence or inference confidence where applicable.
ConfirmationStatus	suggested / confirmed / rejected. Manual entries may start confirmed.
EvidenceRef	Optional link to transcript segment, note, product interaction or other supporting evidence.

Amount semantics: Keep at least three monetary meanings separate: Need Potential (approximate need value), Opportunity Forecast/Expected Value (commercial estimate), and Quote Total (actual commercial offer). They must never be overloaded into one field.

7. AI-Assisted Need Detection

AI is an assistant to the CRM domain. It may infer needs from evidence, but it must not bypass domain commands, authorization, tenant policy or evidence requirements.



1. Ingest evidence through approved application services; do not let the agent query production tables directly.
2. AI produces structured NeedSuggestion objects, not direct database mutations.
3. Each suggestion includes source, confidence and evidence reference.
4. Default behavior: user confirms/rejects suggestion. Tenant policy may permit auto-confirm only for low-risk cases and explicit thresholds.
5. Confirmed needs update Need Potential through domain logic.
6. Every AI-originated change remains traceable in activity/evidence history.

AI boundary: AI must not independently mark an Opportunity Won/Lost, send a formal document, grant consent, or change legal/commercial truth unless an explicit tenant policy and authorization path permits that action.

8. Quote and Customer Portal - Sales Boundary

Quote is deliberately outside CRM. CRM owns the Opportunity; Sales owns the Quote. The two are linked by reference and events.

```

CRM / Opportunity (Open)
├── Stage: any tenant-configured Open stage
├── quote.send capability allowed?
│   └── yes
│       └── Sales / Quote
│           └── Public/Secure Quote Link
│               └── Customer Portal
│                   └── View / Download / Comment
│                       └── Accept / Reject / Revision request

```

Quote concern	Target rule
Creation/sending	Allowed in any Open stage where tenant policy permits. Never bound to a hardcoded "Quote Given" stage.
Versioning	A quote is versioned. Customer interaction must identify the exact version viewed/accepted/rejected.
Link security	Opaque, non-guessable token; revocable; optionally expiring; no PII in URL; rate limited; optional OTP for sensitive cases.
Viewed tracking	QuoteViewed event/timestamp. Viewing is an interaction fact, not a lifecycle transition and not consent.
Accepted	QuoteAccepted is a Sales fact. It may trigger WinOpportunity only through tenant policy/automation if acceptance constitutes a commercial win.
Rejected	QuoteRejected does not automatically mean Opportunity Lost; tenant may revise/resend another quote.
Comments	Quote interaction/comments belong to the quote/customer-portal interaction stream; they can also project into CRM activity timeline.

9. Won and Lost Behavior

State	What happens	What does NOT happen
Won	Opportunity is positively closed. Downstream order/contract/delivery/invoice/document flows may start. Feedback may be requested.	CRM does not become the owner of invoice/order/document models. New quote is not normally created on the closed opportunity.
Lost	Opportunity is negatively closed. Structured loss reason, competitor and notes may be recorded. Follow-up/new opportunity may be created.	No normal quote send or forward-selling mutation on the closed opportunity.

Reopen: Default behavior is terminal. If a tenant needs reopen, implement an explicit ReopenOpportunity command with permission, reason, evidence and state-transition event. Do not silently flip status fields.

10. Activity Timeline

The CRM timeline should show human actions and important cross-domain events without forcing all of them into the Opportunity aggregate.

Activity/event	Owner/source
Call, meeting, task, note, email	CRM / Work management
Need suggested / confirmed / rejected	CRM + AI
Pipeline stage changed	CRM
Quote created/sent/viewed/downloaded	Sales / Customer Portal projected into CRM timeline
Quote accepted/rejected/revision requested	Sales / Customer Portal projected into CRM timeline
Feedback requested/submitted	Feedback capability
Consent granted/withdrawn	Privacy/Consent capability, optionally summarized in CRM timeline

11. Feedback and Consent

The legacy features should remain, but they must not be mixed into Opportunity status or Quote state.

Capability	Examples	Rule
Feedback	Employee rating, branch rating, tenant rating, service/delivery rating	Generic FeedbackRequest/FeedbackResponse capability linked to Party and related business entity.
Consent	Marketing SMS, email, call, data-processing permission	Purpose-specific ConsentRecord with notice/version, source, granted/withdrawn timestamps and evidence.
Quote interaction	View, download, comment, accept, reject	Commercial interaction; not consent and not generic feedback.

Privacy distinction: A customer opening a quote link is an interaction event. It must never be treated as KVKK/marketing consent.

12. Domain Ownership - Non-Negotiable Boundary

Concern	Owner
Party/customer relationship	CRM / shared Party core
Customer Need	CRM
Opportunity	CRM
Pipeline	CRM
Activities / CRM timeline	CRM projection / Work capability
Quote, QuoteVersion, QuoteLine, pricing/discount proposal	Sales
Order / OrderLine	Sales / Order Management
Invoice	Finance / Billing
Formal document storage	Document platform
Quote public link / portal interaction	Sales / Customer Portal
Feedback/evaluations	Feedback capability
Consent/KVKK permissions	Privacy/Consent capability
AI inference/orchestration	AI Agent Runtime; uses application APIs/tools, not direct DB writes

13. Suggested Target Data Model (Conceptual, Not Final DDL)

Aggregate / table	Important fields
opportunities	id, tenant_id, party_id, lifecycle_status, pipeline_definition_version_id, pipeline_stage_id, owner_ref, currency, forecast_amount?, need_potential_amount?, opened_at, won_at, lost_at, row_version, timestamps
pipeline_definitions	id, tenant_id, name, sector_template_ref?, active_version
pipeline_definition_versions	id, pipeline_id, version, effective_at, status
pipeline_stages	id, pipeline_version_id, code/stable key, display_name, sort_order, active, capability/policy refs
customer_need_definitions	id, tenant_id, stable key, display_name, reference_value, currency, source template, active
opportunity_needs	tenant_id, opportunity_id, need_id, estimated_value_snapshot, source, confidence, confirmation_status, evidence_ref, timestamps
activities / timeline projection	tenant_id, party/opportunity ref, activity_type, source domain, source

	event/ref, actor, occurred_at, metadata
quotes (Sales)	id, tenant_id, opportunity_ref, version, status, currency, totals, validity, timestamps
quote_public_links	quote_version_ref, token_hash, expires_at, revoked_at, access policy
feedback_requests/responses	subject, related entity ref, questions, ratings/comments, timestamps
consent_records	party_ref, purpose, notice/version, status, source, granted_at, withdrawn_at, evidence_ref

14. Required Application Commands and Events

Command / use case	Key invariants
CreateOpportunity	Tenant/party valid; starts Draft; pipeline version selected but Open stage not entered until activation.
OpenOpportunity	Moves Draft → Open and assigns initial Open stage.
ChangePipelineStage	Only while Open; target stage belongs to same assigned pipeline version; authorization required.
Add/Confirm/RejectOpportunityNeed	Need belongs to tenant; snapshot value; preserve provenance/evidence.
WinOpportunity	Only while Open; explicit authority/policy; reason/evidence optional by policy.
LoseOpportunity	Open (or Draft discard if retained); structured loss reason required by tenant policy.
ReopenOpportunity	Optional policy-controlled command; reason + evidence; creates explicit lifecycle event.
CreateQuote / SendQuote	Sales command; requires Opportunity Open and current stage/policy capability.
RecordQuoteInteraction	Sales/Portal event; exact quote version; idempotent.

Event	Why it exists
OpportunityCreated / Opened / StageChanged / Won / Lost / Reopened	Lifecycle and process integration.
OpportunityNeedSuggested / Confirmed / Rejected	AI/human need provenance and analytics.
QuoteCreated / Sent / Viewed / Downloaded / Accepted / Rejected / RevisionRequested	Sales and portal behavior projected into CRM timeline and automation.
FeedbackRequested / Submitted	Experience and performance analytics.

ConsentGranted / Withdrawn	Privacy-state integration and evidence.
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15. Sector Template + Tenant Override Model

Sector templates are accelerators, not runtime hardcoding. A tenant receives a default configuration and may override it.

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Sector Template: Furniture Retail
├─ Default Open pipeline stages
├─ Default Customer Need catalog
├─ Default lost reasons
├─ Default quote-stage capabilities
├─ Default feedback form
├─ AI inference taxonomy/prompts/policies
│   └─ clone / reference
Tenant Configuration
├─ Rename/reorder/disable stages
├─ Change need reference values
├─ Add tenant-specific needs
├─ Change allowed quote stages
└─ Configure automation/policies

```

Versioning rule: Changing a tenant pipeline must create a new configuration version. Existing Open opportunities must not jump stages automatically. Any migration/remapping must be explicit and auditable.

16. Worked Example A - Furniture Store

Lifecycle: Draft → Open → Won/Lost

Open stages:

1. İhtiyaç Analizi
2. Teklif Verildi
3. Pazarlık

Need catalog examples:

```

Yatak Odası    120k
Yemek Odası    90k
Yatak          30k

```

Scenario:

- Salesperson creates Draft opportunity.
- Voice transcript suggests Yatak Odası + Yatak.
- User confirms both needs → Need Potential ≈ 150k.
- Opportunity opened at İhtiyaç Analizi.
- Quote may already be created here if tenant policy allows.
- Quote sent in Teklif Verildi; customer opens link → QuoteViewed event.
- Customer comments and asks revision; Opportunity stays Open.
- Revised quote accepted.
- Tenant automation may call WinOpportunity if accepted quote means confirmed sale.
- Won then allows order/contract/invoice/document flow outside CRM.

17. Worked Example B - Technical Service / Maintenance

Lifecycle: Draft → Open → Won/Lost

Open stages:

1. Talep Alındı
2. Keşif
3. Teklif
4. Onay Bekleniyor

Scenario:

- Customer requests generator maintenance.
- Opportunity opens at Talep Alındı.
- Technician completes Keşif; notes and product/service items infer needs.
- Quote sending is prohibited before Keşif by tenant policy.
- At Teklif stage, Sales sends secure quote link.
- Quote rejected does not auto-Lost; salesperson can revise.
- If contract approval arrives, WinOpportunity executes.
- Won triggers service order/contract handoff.

18. Worked Example C - Simple Retail / Small Business

Lifecycle: Draft → Open → Won/Lost

Open stages:

1. Bekliyor

Scenario:

- Tenant does not need a complex funnel.
- One Open stage is enough.
- Quote capability may be disabled entirely, or allowed at Bekliyor.
- Direct sale can execute WinOpportunity without a quote.
- Same reporting still works because canonical lifecycle is unchanged.

19. Worked Example D - B2B Corporate Sale

Lifecycle: Draft → Open → Won/Lost

Open stages:

1. Discovery
2. Solution Fit
3. Commercial Proposal
4. Procurement Review
5. Negotiation

Scenario:

- Account has multiple contacts/stakeholders.
- Needs are derived from discovery calls and uploaded requirements.
- Multiple quote versions exist.
- QuoteAccepted does not automatically mean Won because procurement/legal approval is still required.
- WinOpportunity occurs only after required approval/contract condition is met.

20. Current State → Target Delta

Baseline source: CRM_CURRENT_STATE_ANALYSIS.md, main commit fd72a51, 2026-09-16. The current module has a strong safety foundation but a narrow Opportunity-centric pilot slice.

Current	Target change
OpportunityStatus: waiting/offered/completed/canceled	Replace/refactor toward lifecycle_status: Draft/Open/Won/Lost plus separate configurable pipeline_stage_id.
Waiting used as draft	Map to Draft semantics.

Offered	Map to Open lifecycle + an appropriate tenant stage (for legacy furniture, "Teklif Verildi"). Do not keep Offered as canonical lifecycle.
Completed	Map to Won.
Canceled	Map to Lost with structured loss reason migration where possible.
Opportunity carries quote/order-shaped fields	Gradually separate Sales Quote/QuoteLine/Order concepts. Preserve compatibility during migration; do not big-bang rewrite without a plan.
CustomerNeed / OpportunityNeed persisted but behavior incomplete	Promote to first-class capability with reference values, snapshot values, source, confidence, confirmation status, AI suggestion flow.
Free-text cancel reason	Introduce tenant/sector lost-reason taxonomy plus optional free-text note.
No activities	Add activity/timeline capability and project cross-domain quote events into it.
No authorization gate	Access/Authorization must guard commands before production.

Migration safety: Do not implement this as a destructive schema rewrite in one step. The coding agent must first inventory current usages, prepare backward-compatible migrations/adapters, and define data migration rules for existing tenants.

21. Required Implementation Sequence

Phase	Deliverable
Phase 0 - Inspect and Plan	Read current CRM/Sales code, tests and migrations. Produce a delta plan, dependency impact, migration plan and test plan. Do not change code until the plan is explicit.
Phase 1 - Lifecycle/Pipeline foundation	Introduce canonical lifecycle and tenant pipeline definitions/versions/stages. Preserve current behavior with compatibility mapping.
Phase 2 - Opportunity commands/API	Create/Open/ChangeStage/Win/Lose/Reopen(if enabled), authorization, idempotency, evidence, outbox, concurrency, RLS.
Phase 3 - Customer Need capability	Need definitions, opportunity need snapshots, source/confidence/confirmation, potential calculation and tests.
Phase 4 - Activity timeline	CRM activity objects/projection and event ingestion.
Phase 5 - Sales Quote split	Introduce Sales Quote/QuoteVersion/QuoteLine and opportunity reference. Move quote-specific behavior without breaking CRM history.
Phase 6 - Customer Portal	Secure quote links, view/download/comment/accept/reject/revision interactions and

	event stream.
Phase 7 - AI inference	Structured NeedSuggestion flow from approved sources/tools; human/policy confirmation; provenance/evidence.
Phase 8 - Feedback/Consent integration	Link existing/new feedback and privacy capabilities through stable references/events.
Phase 9 - Sector templates	Template catalog and tenant overrides/versioning after the underlying generic capabilities are stable.

22. Claude/Codex Guardrails - MUST / MUST NOT

MUST	MUST NOT
Keep canonical lifecycle fixed: Draft/Open/Won/Lost.	Do not make lifecycle tenant-configurable.
Keep pipeline stages tenant/sector configurable and versioned.	Do not hardcode stage names in business logic.
Keep Quote in Sales and link it to Opportunity.	Do not put Quote state or customer-portal state into OpportunityStatus.
Allow quote create/send in any Open stage permitted by policy.	Do not require a specific "Teklif Verildi" stage.
Preserve Need source/confidence/evidence and snapshot value.	Do not let AI silently overwrite confirmed needs or revenue values.
Use domain commands and authorization for mutations.	Do not let AI/portal/integration write tables directly.
Preserve RLS, tenant-safe FKs, concurrency, idempotency, evidence and outbox guarantees.	Do not weaken existing enterprise safety controls for convenience.
Use explicit events for cross-domain integration.	Do not create cross-domain database FKs or reach into another module repository.
Version tenant pipeline configuration.	Do not silently remap existing open opportunities when stages change.
Separate Need Potential, Forecast Amount and Quote Total.	Do not overload one money field with multiple meanings.

23. Acceptance Criteria

- A tenant can operate with 1, 2, 3 or more Open pipeline stages without code changes.
- Two tenants in the same sector may use different stage names/order/capabilities.
- Changing a pipeline creates a new version and does not silently alter existing open opportunities.
- Quote can be sent from any Open stage permitted by tenant policy; no stage-name string is checked in domain logic.
- Quote Viewed/Accepted/Rejected does not directly mutate Opportunity lifecycle unless an explicit automation/policy command does so.
- AI can propose needs from voice/products/notes and every suggestion carries source + confidence + evidence reference.
- Confirmed needs produce a stable Need Potential using value snapshots; later catalog price changes do not rewrite history.

- Need Potential, Opportunity Forecast and Quote Total remain distinct in storage/API/reporting.
- Won/Lost remain canonical and reportable across all sectors/tenants.
- Lost requires structured reason policy where configured and blocks normal new quote sending.
- Won may start downstream formal document/order/finance flows without moving those models into CRM.
- Customer feedback and consent remain separate capabilities and are linked by references/events only.
- All state-changing commands are tenant-scoped, authorized, concurrency-safe, idempotent where externally retryable, evidence-aware and outbox-integrated.
- All new persistence continues to enforce RLS with a non-superuser runtime role and tenant-safe referential integrity.

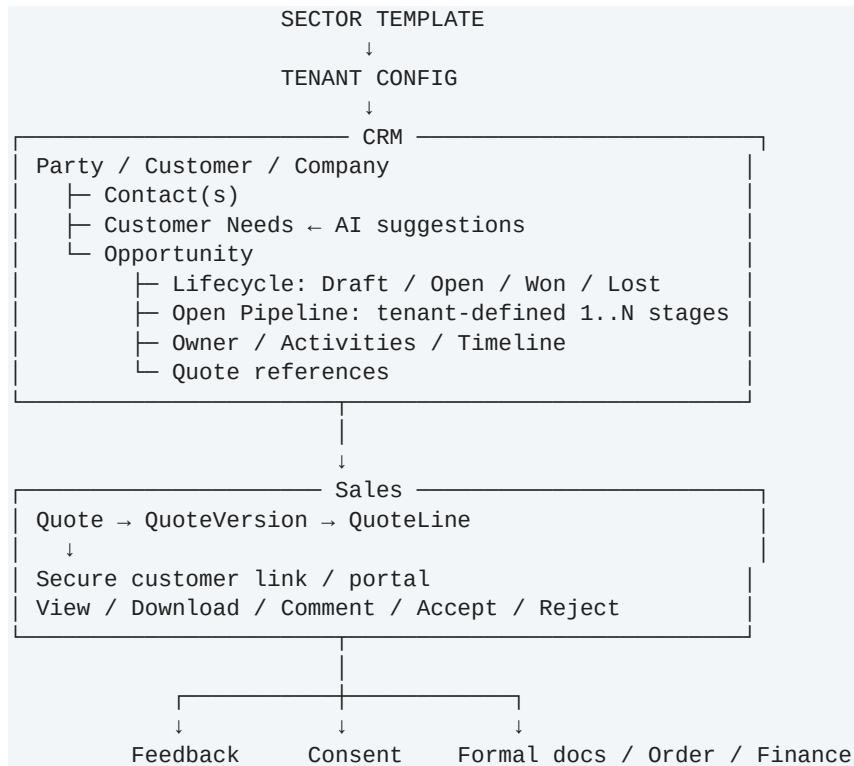
24. Execution Protocol for Coding Agent

When this PDF is given to Claude/Codex, the agent must follow this protocol before making changes:

1. Inventory current code and identify exact files/entities/commands/migrations affected.
2. Produce a decision matrix: Current behavior → Target behavior → Migration strategy → Compatibility risk.
3. List all assumptions. If this PDF does not define a decision, stop and ask; do not invent product behavior.
4. Prepare schema/domain/application/API/event/test changes in dependency order.
5. Preserve existing tests and add new tests for every lifecycle invariant, pipeline rule, tenant boundary and quote-policy boundary.
6. Implement in small reversible increments; each migration must be backward-compatible where feasible.
7. After each phase: build, format, run unit/integration/architecture tests and summarize changed behavior.
8. Do not start the next major phase until the current phase acceptance criteria pass.

Stop condition: If the implementation requires changing any frozen decision in Section 2, the agent must stop and present the conflict instead of changing the model.

25. Final Target Model - One-Page Mental Model



Key rule:

Lifecycle is stable. Pipeline is configurable. Quote is Sales.
Need intelligence is CRM. AI suggests; domain/policy decides.

Appendix A. Current-State Source Note

This specification was prepared against the current-state analysis supplied by the project:

CRM_CURRENT_STATE_ANALYSIS.md, snapshot date 2026-09-16, main commit fd72a51. That analysis reports the current module as a pilot-sized Opportunity/Party slice with strong tenancy/safety controls, CustomerNeed/OpportunityNeed persistence, and a merged CRM/Sales pilot exception. This document defines the target model and therefore intentionally supersedes the pilot-only domain boundaries where explicitly stated above.